NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 1

CANDIDATE NAME

CLASS

BIOLOGY

8875/02

Paper 2 Core Paper

15 September 2014

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do no use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	**15**
2	**7**
3	**10**
4	**8**
Section B	**20**
Total	**60**

This document consists of **20** printed pages and **no** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

1 **Fig. 1.1** shows the effect of increasing substrate concentration on the rate of a particular reaction in the presence and absence of an enzyme.

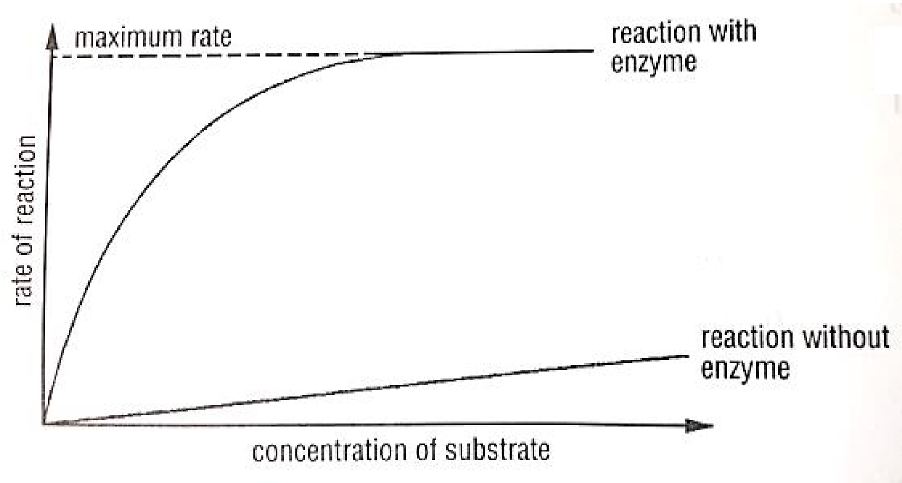

Fig. 1.1

(a) With reference to **Fig. 1.1**, explain the difference between the two graphs.

Higher rate of reaction for reaction with enzyme than reaction without enzyme;

Enzyme **lowers activation energy** of the reaction;

E.g. allowing close proximity of reactants, ensuring correct orientation of reactants, therefore increasing rate of reaction;

Rate of reaction with enzyme eventually **plateaus** whereas for reaction without enzyme there is **no plateau in rate** and rate of reaction **increases steadily** as substrate increases;

Reaction with enzyme: **substrate concentration no longer the limiting factor** + **Enzyme concentration** eventually becomes the limiting factor, with all available active sites occupied at any one time;

For the reaction with enzyme, as substrate concentration increases, the rate of reaction increases due to **increase in number of effective collisions between substrates and active site of the enzymes.** Therefore this increases the number of enzyme-substrate complex formed per unit time and increase in the rate of reaction;

[3]

(b) On **Fig. 1.1**, draw **two** labelled curves to show the effect on the rate of the enzyme catalysed reaction of the addition of

(i) a competitive inhibitor

(ii) a non-competitive inhibitor

[2]

(c) Explain the effect of a competitive inhibitor on the rate of enzyme activity.

Competitive inhibitors are **structurally similar (in terms of shape, size, charge and orientation) to that of the substrate molecule**;

Competes with the substrate to **bind to the active site** of the enzyme, **reducing the number of active sites available** for the substrates to bind and form enzyme-substrate (E-S) complexes;

When substrate concentration is low, it is more likely for the enzyme to collide with competitive inhibitor molecules and form enzyme-inhibitor (E-I) complex;

Thus, the number of active sites available for substrate molecules will be reduced and the rate of product formation will decrease;

When substrate concentration is high, it is more likely for the enzyme molecules to collide with substrate molecules and form E-S complex. Thus, the competitive inhibitor has no effect on rate of reaction.

Learning point: V_{max} of reaction with competitive inhibitor same as V_{max} of reaction without enzyme. K_m of enzyme will increase due to decreased affinity of enzyme active site for substrate molecule.

[3]

(d) **Fig. 1.2** below shows an electron micrograph of a cell from the lung that is specialized to take up material by endocytosis. Darkly staining vesicles containing inorganic aluminium silicate crystals are labeled in the figure. These crystals are only ever seen in such cells in the lungs of cigarette smokers. Aluminium silicate is known to be present in tobacco smoke.

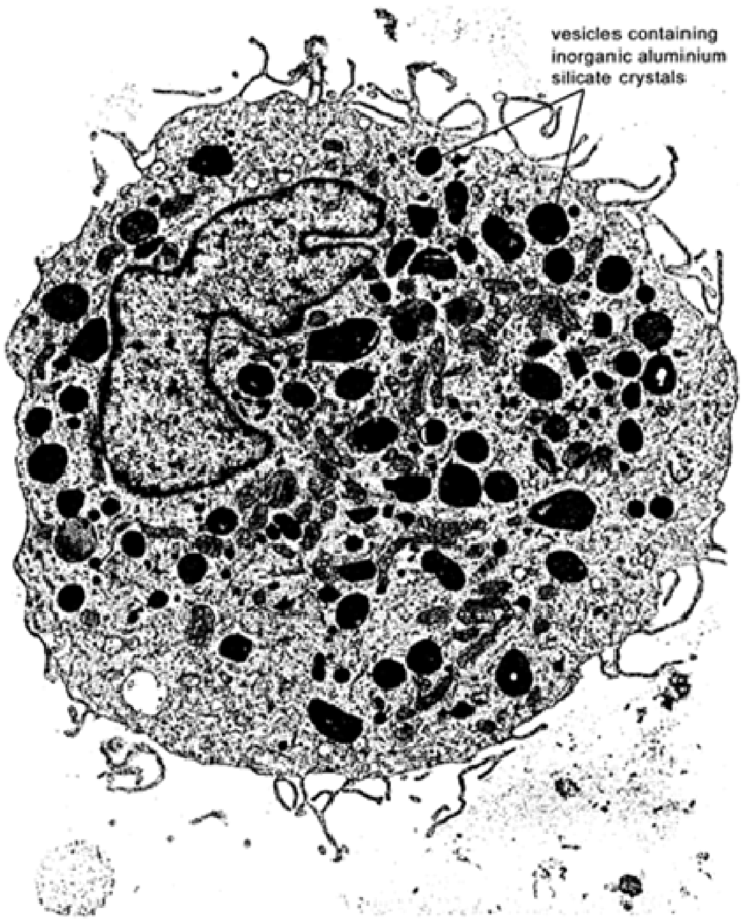

Fig. 1.2

(i) Explain how endocytosis allows foreign material such as aluminium silicate to be taken up by cells.

By the process of **phagocytosis**, whereby there is an extension of **pseudopodia** to **engulf** the particulate material from outside the cell;

When **cell surface membrane fuses**, the membrane pinches off, forming **phagocytic vesicle / endocytic vesicle** containing the aluminum silicate;

[2]

(ii) Describe how foreign material is normally dealt with once in these vesicles.

The **lysosome** containing **hydrolytic enzymes fuses** with the phagocytic vesicle / vesicle containing engulfed material;

Releasing **hydrolytic enzymes** into the vesicle, **hydrolyse** the particles into soluble products;

Useful **soluble** products are absorbed into the cytoplasm and the unwanted **insoluble** products are released into the external medium by **exocytosis**;

[3]

(iii) Explain the function of the numerous mitochondria that are clearly visible in the cell.

Mitochondria is the **site of cellular respiration** where **ATP synthesis** occurs;
necessary to **produce the ATP** required by the cell to carry out endocytosis which is an **active/ energy consuming process;**

[2]

[Total: 15]

2 A pure-breeding variety of tomato plant, variety **A**, produced red fruit which remained green at their bases even when ripe.

Plants of variety **A** were crossed with another pure-breeding variety, **B**, with orange fruit which have no green bases when ripe. The F_1 generation plants all had red fruit with green bases.

(a) Describe the interaction of the alleles,

(i) at the locus **G**/**g**, controlling green-based or not green-based fruit

Green-based fruit is dominant/ G dominant to g;

[1]

(ii) at the locus **R**/**r**, controlling red or orange fruit.

Red fruit is dominant to orange fruit/ R dominant to r;

[1]

(b) Using the symbols given in **(a)**, state the genotype of variety **B.**

rrgg/ ggrr;

[1]

(c) Plants from the F_1 generation were crossed to variety **B** and the F_2 offspring were recorded.

red fruit, green base	53
red fruit, non-green base	58
orange fruit, green base	52
orange fruit, non-green base	55

Using the symbols given in **(a)**, draw a genetic diagram to show that the expected phenotypic ratio of offspring phenotypes is 1:1:1:1.

Parental phenotypes red fruit with green base X orange fruit with non-green base

Parental genotypes Rr Gg X rrgg;

Meiosis

Gametes R Rg rG rg rg ;

Offspring genotypes	RrGg	Rrgg	rrGg	rrgg	;
Offspring phenotypes	red fruit green base	red fruit non green base	orange fruit green base	orange fruit non green base	
Phenotypic ratio	1 :	1 :	1 :	1 ;	

[4]

[Total: 7]

3 Figure **3.1** shows the main stages in the polymerase chain reaction (PCR).

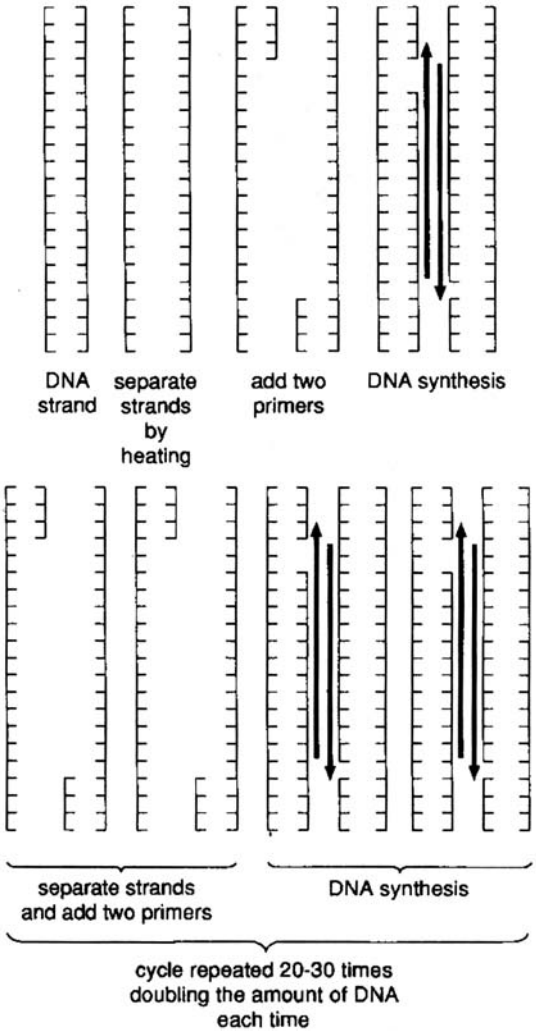

Fig. 3.1

This technique allows multiple copies of a selected length of DNA to be made in the laboratory. Portions of the base sequences at either end of the DNA to be copied are used to produce to short lengths of synthetic DNA, one complementary to each strand of the DNA double helix. These short lengths of DNA serve as primers for *in vitro* DNA synthesis, which is catalysed by DNA polymerase, and they determine the ends of the final DNA fragment that is obtained.

Each cycle of the reaction requires a brief heat treatment to separate the two strands of the DNA double helix. A subsequent cooling of the DNA in the presence of a large excess of the two primers allows their specific attachment to their complementary DNA sequences. The mixture is then incubated with DNA polymerase and the four different DNA nucleotides so that the regions of DNA, starting from the primers, are selectively synthesised.

When first developed, PCR was slow and laborious because a crucial enzyme for the process, DNA polymerase, was denatured at high temperature and has to be added to each cycle. However the introduction of enzyme, *Taq* DNA polymerase (isolated from the heat tolerant bacterium, *Thermus aquaticus*) revolutionised PCR so that the technique could be fully automated.

(a) (i) List the components of a nucleotide.

phosphate group + pentose sugar (deoxyribose sugar) + nitrogenous base;

[1]

(ii) Explain what is meant by *four different nucleotides*.

four different nitrogenous bases attached to pentose sugar: adenine, guanine, cytosine, thymine;

[1]

(b) Suggest why the brief heat treatment separates the two strands of the DNA double helix.

two strands of DNA joined by H-bonds between A=T and C≡G; [students write in full]

at high temperature, molecule has high/increased kinetic energy, excessive/increased vibrations causes weak H-bonds to break (or) H-bonds easily broken by heat treatment;

[2]

(c) Suggest why DNA synthesis takes place in opposite directions on the two separated strands of the DNA double helix.

two DNA strands are *antiparallel* 5'→ 3' and 3'→5';

new DNA strand syn from *5'→3' direction*;

DNA polymerase only adds nucleotide to 3' end of growing polynucleotide due to specific shape, charge, orientation; [note: *DNA polymerase only work in this direction]*

[2]

(d) Explain the advantages of using *Taq* polymerase.

PCR performed at high temp and Taq is a heat resistant/thermostable DNA polymerase;

does not get denatured by high temperature like other normal polymerases;

PCR reaction can continue for many cycles without having to replace the enzyme;

Taq polymerase synthesised commercially (in large amounts) via genetic engineering;

[2]

(e) State the limitations of PCR.

Some knowledge of DNA/amino acid sequence of desired gene/protein needed to synthesise flanking nucleotide primers;

many sequences in starting DNA sequence may be complementary to the short primers, results in amplification of non-target DNA sequence;

Taq polymerase does not perform proofreading/ mistakes in complementarity may occur when amplified;

[2]

[Total: 10]

4 In 1999, genetically modified maize was widely grown in the maize-growing areas of the USA. One of the genetically modified varieties of maize contains a gene (*Bt*) from a bacterium, *Bacillus thuringiensis*. The gene codes for a toxin, which is expressed in the leaves and acts as an insecticide.

(a) Explain briefly

(i) what is meant by a **genetically modified** variety of maize;

Genetic material of the maize is altered by the insertion of a **transgene**, obtained from a **different species** and **inserted into the maize genome**;

Gene can be inherited to subsequent generations as part of the plant genome and can be **expressed in the phenotype** of the maize;

[1]

(ii) why farmers in the USA grow maize carrying the *Bt* gene.

Maize with Bt gene express Bt protoxin in the plant, which kills the insects when ingested by causing gut cells of insects to lyse;

Increases the yield, therefore higher economic returns and less losses due to insect damage;

Lesser pesticides needed, therefore higher economic savings;

[2]

The *Bt* gene is also expressed in pollen, which may be dispersed at least 60m away from the parent plant, by the wind. In the USA, milkweed frequently grows around the edge of maize fields and is fed upon by caterpillars of the Monarch butterfly.

In an experiment into the environmental effects of *Bt* maize, leaves of milkweed were divided into three groups, **A**, **B** and **C**, and treated in the following ways:

A dusted with pollen from genetically modified maize carrying the *Bt* gene;
B dusted with pollen from maize that had not been genetically modified;
C not dusted with pollen.

Monarch caterpillars were then placed on the leaves and the mean area of leaf eaten per caterpillar and the survival of the caterpillars was measured over four days. The results of the experiment are shown in **Fig**. **4.2** and **4.3**.

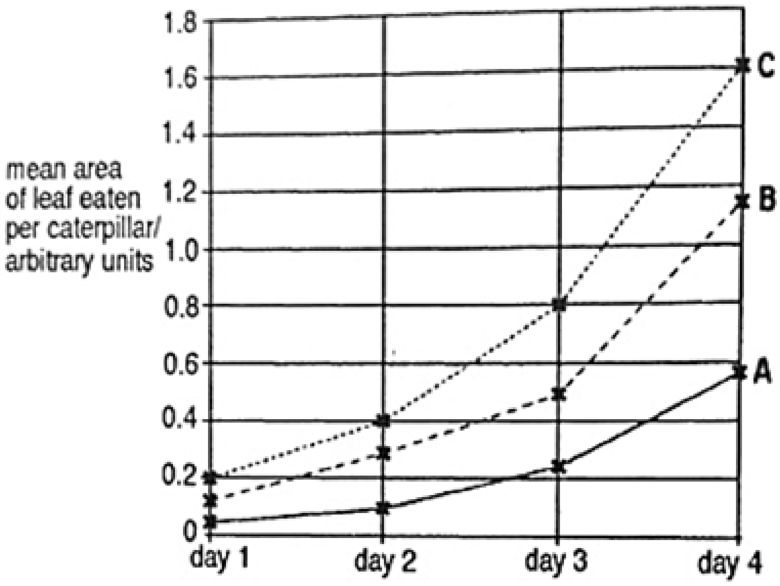

Fig. 4.2

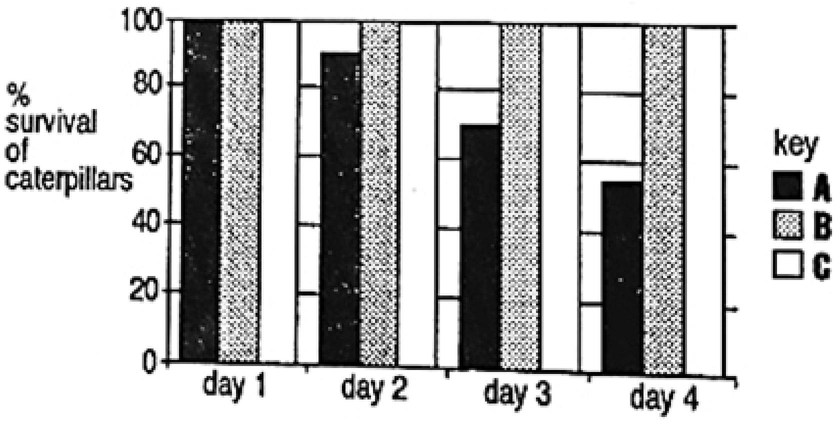

Fig. 4.3

(f) With reference to **Fig. 4.2** and **4.3**, comment on the effect of eating pollen from different maize plants on

(i) the mean area of leaf eaten per caterpillar

Mean area of leaf eaten per caterpillar **increases** for all the 3 groups of milkweed leaves from Day 1 to Day 4;

The **lowest mean area** of leaf eaten per caterpillar for all 4 days was **group A**, which consists of milkweed dusted with pollen from GM maize carrying the *Bt* gene (**0.05, 0.1, 0.25, 0.6 a.u**.). Whereas the **highest mean area** of leaf eaten per caterpillar for all 4 days was **group C**, which consists of milkweed not dusted with pollen (**0.2, 0.4, 0.8, 1.6 a.u.**);

Mean area of leaf eaten per caterpillar was **highest** on **day 4** for all 3 groups, the highest was in **Group C** (1.6 a.u.), followed by **Group B** (1.2 a.u.), then the lowest is **Group A** (0.6 a.u.);

(MAX 1)

Milkweed leaves containing pollen from genetically modified maize carrying the *Bt* gene was least consumed by the caterpillars while leaves containing no pollen was most consumed by the caterpillars;

[2]

(ii) the survival of Monarch butterfly caterpillars

100% survival of caterpillars which fed on leaves without dusted pollen and leaves with pollen from non-GM maize for **all 4 days**;

Survival of caterpillars that fed on leaves dusted with pollen from genetically modified maize carrying the *Bt* gene **decreased steadily** from **day 1 to day 4, (100% to 55%);**

Bt toxin does affect the survival of monarch butterfly caterpillars;

[3]

[Total : 8]

Section B

Answer **ONE** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 (a) Explain how genetic variation can arise. [7]

(b) Explain using two named examples, how environmental forces act as forces of natural selection. [7]

(c) Discuss the benefits and ethical concerns of the human genome project for humans. [6]

[Total: 20]

5 (a) Distinguish between the structure of protein and DNA. [7]

(b) Describe the properties of plasmids that allow them to be used as DNA cloning vectors. [5]

(c) With reference to the term 'limiting factor', discuss the effect of temperature on the rate of photosynthesis. [8]

[Total: 20]

(a) Explain how genetic variation can arise. [7]

1 The formation of **four haploid cells / gametes** during **meiosis** is an important step during sexual reproduction, which results in genetic variation between individuals in a population.

2 Meiosis also results in genetic variation as the reduction division to form haploid gametes allows the **combining of genetic material from two parents / individuals**;

3 During meiosis, **crossing over of non-sister chromatids of homologous chromosomes**, at the chiasmata, occurs during **prophase I**,

4 allowing **corresponding sections to be exchanged**, creating **new combination of alleles** in each chromatid;

5 **Independent assortment of homologous chromosomes** occurs during **metaphase I**, where the orientation of homologous pair of chromosomes along the metaphase plate is **independent of other bivalents**;

6 This is followed by **independent segregation** during **anaphase I**, resulting in numerous possible **chromosomal combinations /** **new combination of alleles in gametes** in a gamete, i.e. 2^n, where n = number of homologous pairs of chromosomes;

7 In addition, during **fertilization**, **random fusion of gametes** occurs, resulting in numerous combinations of a zygote;

8 Gene mutation is a change in DNA sequence e.g. deletion, insertion, substitution or inversion of one or several nucleotides;

9 which give rise to new alleles and can cause changes to amino acid sequence and structure of proteins leading to different phenotype. Eg: sickle cell anaemia;

10 Chromosomal mutation is a change in structure or number of chromosomes e.g. deletion, inversion, translocation or duplication of a segment of chromosome;

11 give rise to new genotypes due to reshuffling of alleles on a chromosome;

12 Change in number of chromosome e.g. aneuploidy is brought about by non-disjunction giving rise to an extra or lack of one chromosome or e.g. polyploidy give rise to extra sets of chromosomes;

(b) Explain using two named examples, how environmental forces act as forces of natural selection. [7]

Named Example 1: Peppered moths (*Biston betularia*)

1 For natural selection to occur, there must be **heritable variation** for a particular trait. In this case, it is the alleles of the gene which determine the colour of the peppered moth.

2 There are **two types** of peppered moths (*Biston betularia*). The **lighter** form (white with black spots) and the **melanic** form (black with white spots).

3 Before 1848, the lighter form of moths had a **selective advantage** as they were **well-camouflaged** against the background of **light coloured, lichen-covered tree barks**.

4 Since they are less visible to their predators (birds), therefore they were **selected for**, which in turn able to **survive to maturity and reproduce to have viable offspring, thereby passing on their alleles to their offspring.** Therefore white peppered moths were found in **very high frequency.**

5 The **melanic** form is very clearly seen / more visible on the light-coloured tree bark, therefore they were **selected against / have selective disadvantage** as they became **easy prey to the birds**.

6 With the **industrial revolution**, lichens on bark of trees were killed and barks were **covered with soot** & thus **appeared darker**.

7 The **lighter** forms of moths were **selected against** / had **selective disadvantage**, as they became more visible and **easy prey to birds**. Thus, their numbers **declined**.

8 The darker forms of moth however, were **well camouflaged** against the blackened tree bark and thus **proliferated** and **frequency** of melanic moths increased.

9 Hence there was **differential survival and reproductive success** associated with the possession of a particular trait, in this case the colour and form of the peppered moth, leads to a **change in allele frequency in a population** for this trait.

(max 4)

Example 2:
Named example: Galapagos finches

1 For natural selection to occur, there must be **heritable variation** for a particular trait. In this case, it is the alleles of the gene which determine the size and depth of the beaks in Galapagos finches.

2 During wet seasons, there is a **change in abundance of food source available**, whereby now small seeds are abundant.

3 Finches with **small beaks** are at a **selective advantage** as it allows it to feed on small tender seeds.

4 During dry seasons / drought, small seeds are less abundant and only large hard seeds are available.

5 Finches with **large, more powerful beaks** fed on the large hard seeds were **selected for / selective advantage.**

6 The type of beak phenotype that is being selected for depends on the **availability of the food source** due to **different environmental conditions or different habitats.**

7 Individuals which are more adapted to surviving in a particular habitat will **survive to maturity, reproduce to produce viable offspring and pass on the alleles to the next generation.**

8 Hence there was **differential survival and reproductive success** associated with the possession of the particular beak type, therefore this leads to a **change in allele frequency in a population** for beak type.

(max 4)

(c) Discuss the benefits and ethical concerns of the human genome project for humans. [6]

Benefits (Max 4): State + describe one example within category

Molecular Medicine:

Improved diagnosis of disease/
Earlier detection of genetic predispositions to disease/
Rational drug design/
Gene therapy and control systems for drugs/
Pharmacogenomics "custom drugs";

DNA Forensics (Identification)

Identify potential suspects whose DNA may match evidence left at crime scenes;
Exonerate persons wrongly accused of crimes/
Identify crime and catastrophe victims/
Establish paternity and other family relationships/
Identify endangered and protected species as an aid to wildlife officials (could be used for prosecuting poachers)/
Detect bacteria and other organisms that may pollute air, water, soil, and food;
Match organ donors with recipients in transplant programs/
Determine pedigree for seed or livestock breeds/
Authenticate consumables such as caviar and wine;

Risk Assessment

Assess health damage and risks caused by radiation exposure, including low-dose exposures/

Assess health damage and risks caused by exposure to mutagenic chemicals and cancer-causing toxins/

Reduce the likelihood of heritable mutations;

Bioarchaeology, Anthropology, Evolution, and Human Migration

Study evolution through germline mutations in lineages/

Study migration of different population groups based on female genetic inheritance/

Study mutations on the Y chromosome to trace lineage and migration of males/

Compare breakpoints in the evolution of mutations with ages of populations and historical events

Energy and Environmental Applications

Use microbial genomics research to create new energy sources (biofuels)/

Use microbial genomics research to develop environmental monitoring techniques to detect pollutants/

Use microbial genomics research for safe, efficient environmental remediation/

Use microbial genomics research for carbon sequestration;

Agriculture, Livestock Breeding, and Bioprocessing

Disease-, insect-, and drought-resistant crops;

Healthier, more productive, disease-resistant farm animals;

More nutritious produce;

Biopesticides;

Edible vaccines incorporated into food products;

New environmental cleanup uses for plants like tobacco;

Ethical Concerns (Max 4)

Fairness in the use of genetic information by insurers, employers, courts, schools, adoption agencies, and the military, among others.

Privacy and confidentiality of genetic information;

Psychological impact and stigmatization due to an individual's genetic differences;

Reproductive issues including adequate informed consent for complex and potentially controversial procedures / use of genetic information in reproductive decision making / reproductive rights;

Clinical issues including the education of doctors and other health service providers, patients, and the general public in genetic capabilities, scientific limitations, and social risks / implementation of standards and quality-control measures in testing procedures;

Uncertainties associated with gene tests for susceptibilities and complex conditions (e.g., heart disease) linked to multiple genes and gene-environment interactions;

Conceptual and philosophical implications regarding human responsibility, free-will versus genetic determinism / concepts of health and disease;

Health and environmental issues concerning genetically modified foods (GM) and microbes;

Commercialization of products including property rights (patents, copyrights, and trade secrets) and accessibility of data and materials;

(a) Distinguish between the structure of protein and DNA. [7]

	Features	Protein	DNA
1	Overall structure	**Globular proteins** consists **linear polypeptide chains** folded into compact **spherical** shape; Can also be **fibrous proteins**, consisting of linear polypeptide chain to form fibrils, which in turn unite to form fibres**;**	**Linear DNA** consists of a double polynucleotide chains wounded into a **double helix ;**
2	Monomers	Proteins are made up of **amino acids**	DNA are made up of deoxyribo**nucleotides ;**
3	Types of different monomers.	There are **20 different common amino acids** present in proteins	There are **four** different nucleotides adenine, thymine, cytosine, guanine **;** ® uracil
4	Bonds btn monomers	Amino acids are held by **peptide bonds between –COOH & -NH** to form protein	Nucleotides are held by **phosphodiester linkages** between 3' –OH of 1 nt and the phosphate group of another nucleotide**;**
5	Other bonds supporting overall structure	Structure of protein is maintained by **hydrogen bonds**, **ionic bonds**, **disulfide bonds**, **and/or hydrophobic interactions between R- groups;**	The double helix structure of DNA is maintained by **hydrogen bonding between nitrogenous bases** and **hydrophobic interaction between stacked bases ;**
6	Properties of the molecules of the biomolecule	**Hydrophobic amino acids on the interior** and **hydrophilic amino acids on the exterior of protein;**	**Hydrophobic nitrogenous bases on the interior** and **hydrophilic sugar-phosphate backbone on the exterior of protein;**
8	Elements present	some enzymes contain element **sulphur** in some amino acids (e.g. cysteine)	DNA contains element **phosphorus** in phosphate group of the nucleotide**;**

***(10 marking points for this qns)**

(b) With reference to the term 'limiting factor', discuss the effect of temperature on the rate of photosynthesis. [8]

Limiting factor

1 When the rate of photosynthesis is limited by more than one factors, the rate is limited by the factor which is nearest its minimum value;

2 A limiting factor directly affects the rate of reaction if its magnitude is changed

Effect of temperature

3 Affects mainly light-independent stage and, to a certain extent, the light-dependent stage;

4 Because light-independent reactions are enzyme-controlled;

5 Increase in temperature, increases rate of photosynthesis;

6 Rate of photosynthesis doubles for each rise of 10°C until optimum temperature is

7 Reached;

8 Rate of photosynthesis decreases at higher temperature (>40°C) as enzymes start to denature;

9 Relevant diagram;

(c) Describe the properties of plasmids that allow them to be used as DNA cloning vectors. [5]

1 Plasmids are **<u>small circles of DNA</u>** which can enter host cells easily by transformation;

2 Plasmids contain **<u>restriction enzyme recognition sites</u>** that allow insertion of foreign genes using restriction enzymes and ligase;

3 Plasmids usually contain **<u>at least two selection marker genes</u>** (e.g. two antibiotic resistance genes or one antibiotic resistance gene + lac Z gene). Insertional inactivation of selection marker is used to identify recombinant cells;

4 They contains an **<u>origin of replication</u>** which allows them to replicate autonomously / independently of the bacterial chromosome;

5 **<u>High copy number</u>** plasmids are desirable so that the quantity of plasmid DNA that can be purified from each host cell is high;